

Stakeholder Report

Credit Delinquency Risk Analysis

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1. Executive Summary

The objective of this analysis was to identify customers who are at the highest risk of credit card delinquency using customer financial and behavioral data. Based on the exploratory data analysis (EDA) and the proposed predictive modeling approach, several important risk indicators were identified. These insights can help the collections team take proactive actions, improve customer engagement, and reduce future delinquency rates while ensuring responsible and transparent AI-driven decision-making.

2. Predictive Insights

Top 3 Risk Factors

- **High Credit Utilization** — Customers using a large percentage of their available credit may experience repayment difficulties.
- **High Debt-to-Income Ratio** — A higher debt burden relative to income may make timely repayment more difficult.
- **Multiple Missed Payments** — Missed-payment history is an important behavioral signal for future payment risk.

High-Risk Customer Segments: Customers with high credit utilization, high debt-to-income ratios, and repeated missed payments should be prioritized for early intervention such as payment reminders, flexible repayment plans, or financial guidance.

3. Business Recommendation (SMART Goal)

Goal: Within the next **3 months**, identify customers whose **Credit Utilization exceeds 80%** and send personalized payment reminders within **7 days** of crossing this threshold, with the objective of **reducing the credit card delinquency rate by at least 10%**.

Business Rationale: Early identification allows the collections team to take preventive action before customers become delinquent and use collection resources more efficiently.

4. Ethical & Responsible AI Considerations

- **Fairness:** Evaluate model performance across employment groups and other customer segments; investigate significant differences.
- **False Positives:** Monitor False Positive and False Negative rates and use human review for important collection decisions.
- **Explainability:** Communicate predictions in plain language by showing key factors such as credit utilization, missed payments, or debt-to-income ratio.

5. Conclusion

The analysis identified Credit Utilization, Debt-to-Income Ratio, and Missed Payments as important indicators of credit delinquency risk. A Logistic Regression-based prediction approach combined with proactive intervention can help Geldium identify potentially high-risk customers earlier and improve collection efficiency. Fairness monitoring, human oversight, and transparent decision-making support responsible AI use.